

# PRAJWAL KUMAR K

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[prajwal-k-tech.github.io](https://prajwal-k-tech.github.io) | [github.com/Prajwal-k-tech](https://github.com/Prajwal-k-tech)

[codeforces.com/profile/oGhostyyy](https://codeforces.com/profile/oGhostyyy) | [codechef.com/users/oghostyyy](https://codechef.com/users/oghostyyy) | [leetcode.com/u/prajwal\\_k\\_k](https://leetcode.com/u/prajwal_k_k)

## EDUCATION

**Indian Institute of Information Technology, Kottayam**

Kottayam, Kerala

*B.Tech in Computer Science and Engineering; CGPA: 8.66/10.0*

*Aug 2024 – May 2028 (expected)*

## RESEARCH INTERESTS

Algorithms and systems, game theory, machine learning and data science, AI safety, and NLP.

## SKILLS

**Programming:** Python, C++, Rust, MATLAB

**ML and data:** PyTorch, TensorFlow, scikit-learn, pandas, NumPy, Matplotlib

**Research tools:** Jupyter, Simulink, FastAPI, Next.js, SQL

**Current study:** ISLP, Mathematics for Machine Learning, Stanford CS231n (ongoing)

**Languages:** Fluent in English, Kannada, German and Hindi.

## EXPERIENCE AND VOLUNTEERING

**Technical Lead, Openverse (Open Source Club)**

May 2025 – Present

*IIIT Kottayam*

*Kottayam, Kerala*

– Lead technical work and PR reviews; contribute sorting visualizations to the team-built Algorithmia platform.

**Coordinator, Coders Club**

Jan 2025 – Present

*IIIT Kottayam*

*Kottayam, Kerala*

– Curate problems, configure Codeforces Gym rounds and help run campus programming contests.

## RESEARCH AND SELECTED PROJECTS

**PokeForge** | *Gen 9 OU Battle Agents, Python, PyTorch*

GitHub | In progress

– Reproduced pinned FoulPlay and Metamon baselines in isolated simulator environments; testing simulator conformance and agent strength before drawing performance conclusions.

**Bluff** | *Repeated-Play Opponent Modeling*

GitHub

– Built an imperfect-information card-game simulator where agents lie and call bluffs; persistent Bayesian profiles model recurring opponents across rounds.

– Across eight synthetic profiles, removing the learned bluff detector reduced draw-aware score by 4.9–5.8 percentage points; this result is limited to the simulator and tested roster.

**Drishti-XAI** | *Retinal Imaging, MATLAB, Python*

SIH 2026 submission

– Built the MATLAB/Python prototype and evaluation pipeline for explainable retinal screening, covering image-quality checks, grading, segmentation, Grad-CAM, calibration and reporting.

– At a fixed model/threshold, internal APTOS (n=733) achieved 0.908 sensitivity/0.877 specificity; external Messidor-2 achieved 0.363/0.865 with AUC 0.666, exposing a transfer gap.

**KPlaceNet** | *Image Geolocation, PyTorch, ResNet-50*

GitHub

– Evaluated quadtree-based CNN geolocation; the best of 12 ResNet-50 configurations placed 3.33% of 3,000 spatially separated OSV-5M test images within 200 km.

## PUBLICATION AND ACHIEVEMENTS

K. Prajwal Kumar, Rubiya Subair, and Shajulin Benedict. “Collaboration-Enabled Energy-Aware Learning Model Exchange (CE-LME) Framework for Edge Computing-Assisted Applications.” *2025 IEEE 3rd International Conference on Inventive Computing and Informatics*, pp. 1390–1395. DOI. First-listed author; conference presenter. Algorithmia selected for OOSC 4.0; team lightning talk at OOSC 3.0 and presentation at T4E 2025, IIT Madras. Best Software Award, IEEE MACE .hack(); '26, Team Hogorithm.

## OPEN-SOURCE CONTRIBUTION

PettingZoo Leduc Hold'em (PR #1500, open): migrated the environment to an OpenSpiel-backed Shimmy adapter; focused tests passed.